

Chapter 2: Demand and Consumer Choice

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Darren's demand for gasoline

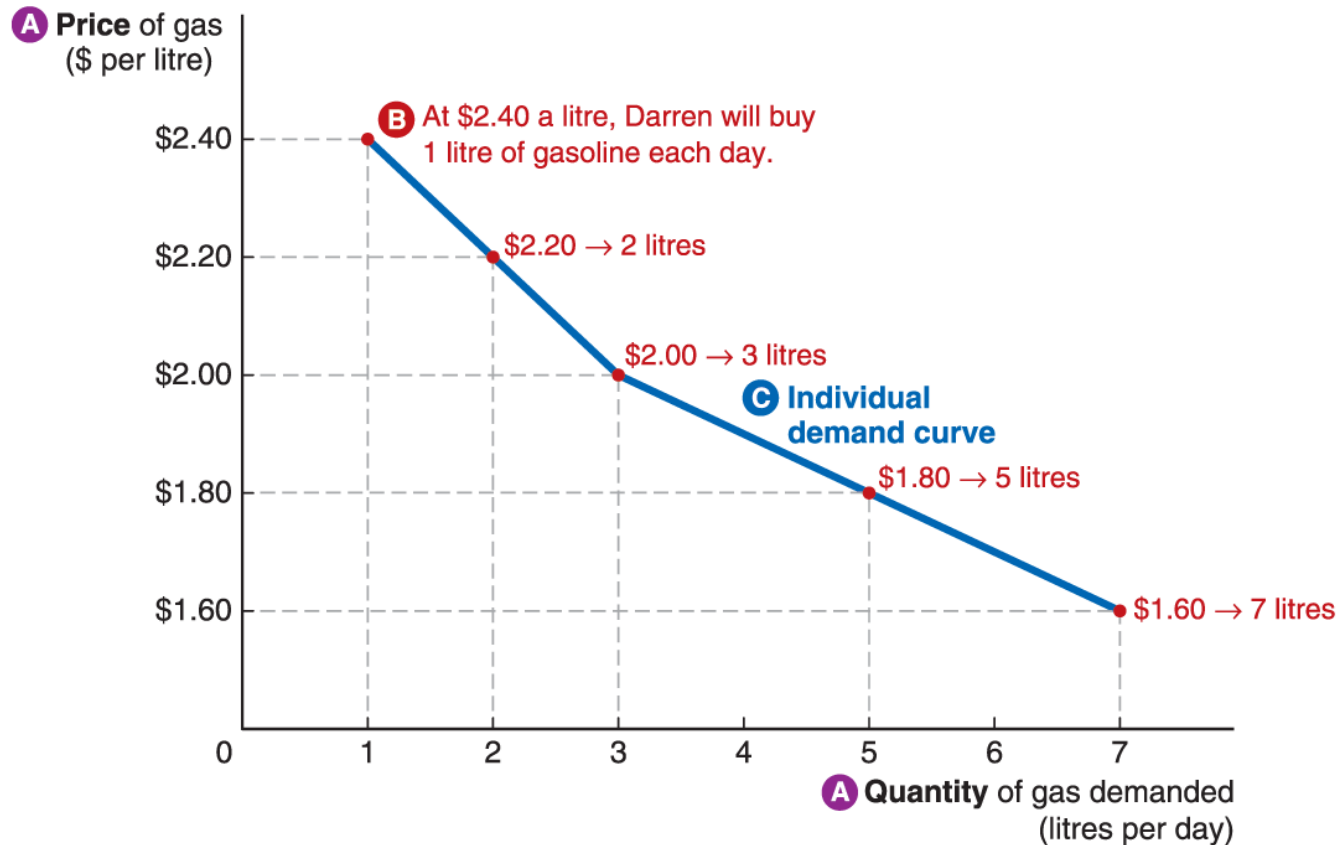
Name: Darren

We are interested in understanding next year's demand for gas.
What quantity of gas do you expect to buy per day, on average:

If the price is \$2.40 per litre?	1 litre
If the price is \$2.20 per litre?	2 litres
If the price is \$2.00 per litre?	3 litres
If the price is \$1.80 per litre?	5 litres
If the price is \$1.60 per litre?	7 litres

- Darren owns a car and uses it to commute to work and school, go shopping, and visit friends and family.
- Darren was surveyed about his demand for gasoline at various hypothetical prices. The chart shows the results.

Darren's individual demand curve



A Price is on the vertical axis and **quantity demanded** is on the horizontal axis.

B When the price is \$2.40 per litre, Darren will use 1 litre of gas per day, but when the price is \$1.80 per litre, he will use 5 litres per day.

C The **individual demand curve** shows the quantity demanded at each price.

Law of demand



- The **law of demand** says that our quantity demanded increases when the price decreases, and vice-versa.
- In other words, individual demand curves are downward sloping.
- The demand curve shows what happens to demand when the price changes **holding everything else constant** (“ceteris paribus”).

What's behind Darren's demand curve?

Priority	Darren's thoughts	Marginal benefit
1 (Highest priority)	If I could only use one litre of gas each day (on average), I'd use it to drive to work. I prefer this to catching the bus. The time and money saved are worth \$2.40 each day to me.	\$2.40
2	If I could use a second litre of gas, I'll also drive to school. I can bike to school instead, but I sometimes arrive all sweaty. Avoiding this by driving is worth \$2.20 per day to me.	\$2.20
3	With a third litre of daily gas I will have enough to do my shopping at No Frills about once a week. The alternative is to shop at my nearby supermarket, which is more expensive. Going to No Frills instead saves me, on average, \$2.00 each day (\$14.00 a week).	\$2.00
4	With a fourth litre of gas, I can drive to hang out with my friends. I could get a ride instead, since all of my buddies live nearby, but it's nice to have the flexibility that driving gives me. I get an average of about \$1.90 per day in benefits from this.	\$1.90
5	A fifth litre allows me to drive to the gym regularly. But I could jog there instead, which is a good warm-up. Saving time is useful, but given that I have to warm up anyway, the benefit of driving to the gym is worth only \$1.80 per day, on average.	\$1.80
6	If I buy a sixth litre of gas, I'll use it to do my errands. But I'm just as happy just walking around town to do these errands. The benefit of driving to do errands is only \$1.70.	\$1.70
7 (Lowest priority)	If I buy a seventh litre, I'll use it to take a scenic drive when I need some quiet time. But I'm just as happy spending quiet time at home, so the benefit of this option is pretty low. Perhaps this litre yields a benefit as small as \$1.60.	\$1.60

If the price of gas is \$1.85 per litre, how much gas should Darren buy each day, on average?

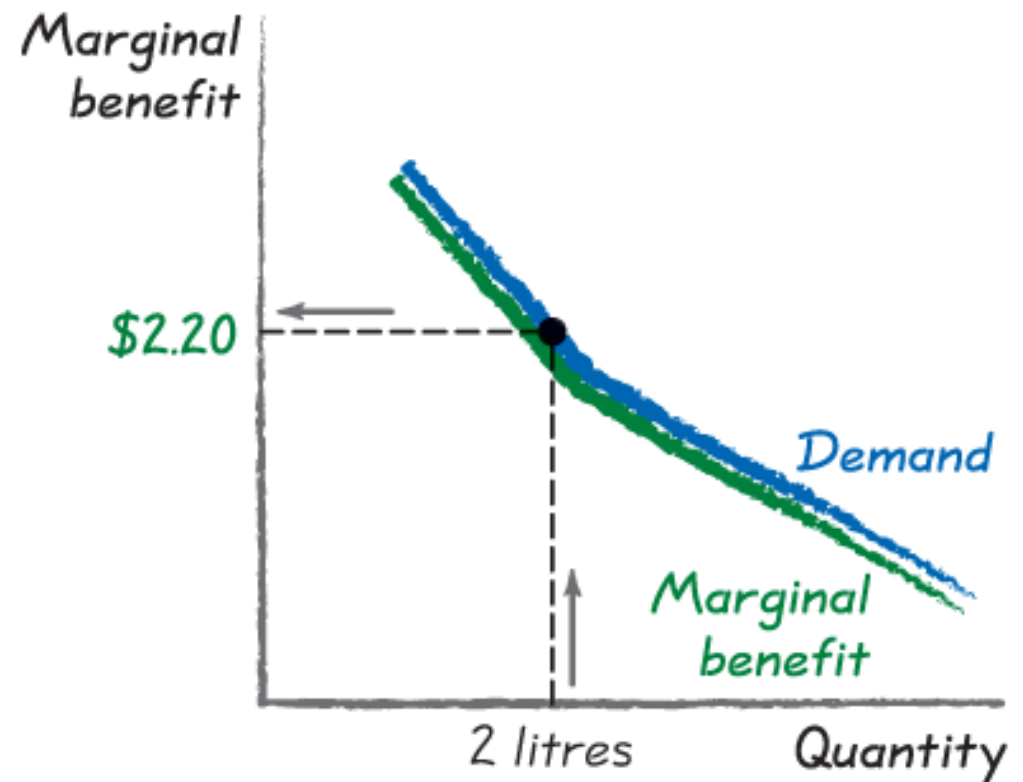
Use the **marginal principle** to get the answer.

The rational rule for buyers

- The rational rule says to keep buying more until the **marginal benefit** equals the **marginal cost**.
- Since the marginal cost is the price of a litre of gas, we should buy until the marginal benefit is equal to the **price**.

$$\text{Price} = \text{Marginal benefit}$$

- The law of demand is a consequence of the **diminishing marginal benefit** of consumption.



From individual to market demand

- From surveying a random sample of 400 consumers, we can estimate the **market demand** for gasoline in the population of 40 million people.

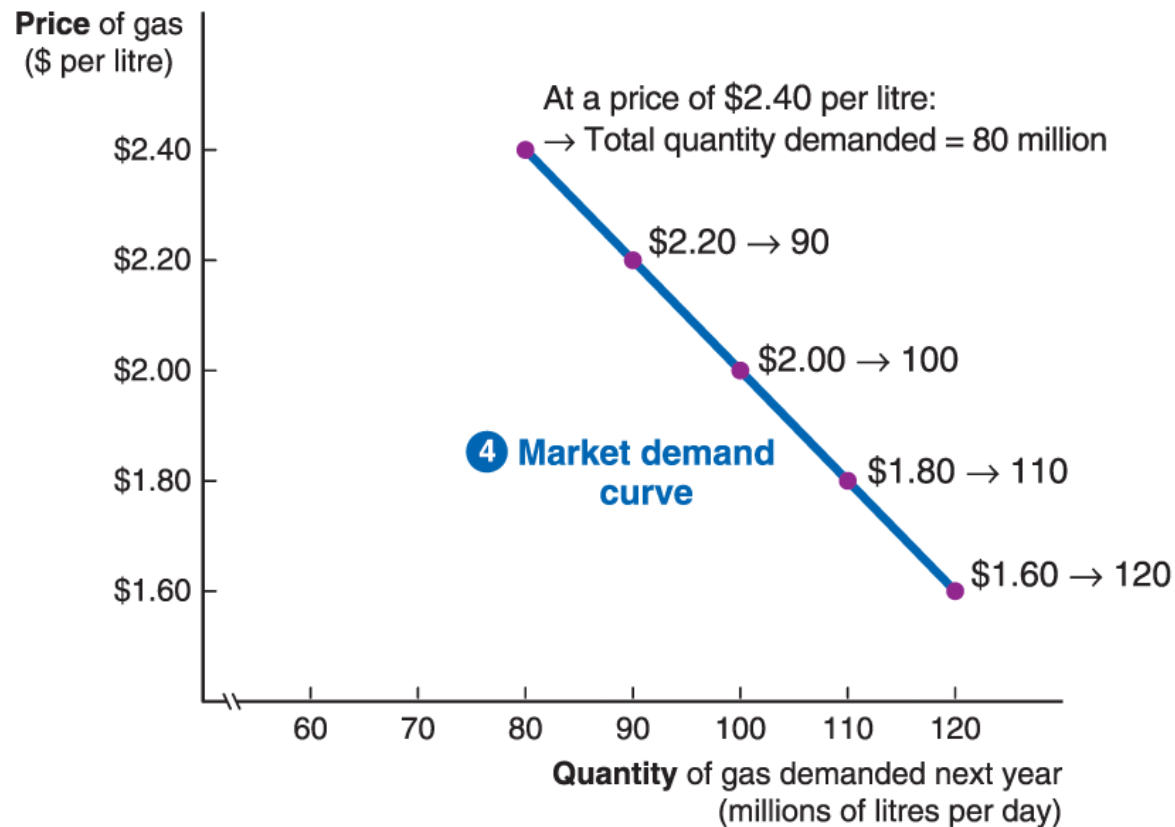
Panel A: Individual Demand

Step 1: Run a survey					
Price (\$ per litre)	Darren's demand		Brooklyn's demand		... 398 other people ...
\$1.60	7	+	4	+	...
\$1.80	5	+	3	+	...
\$2.00	3	+	2	+	...
\$2.20	2	+	1	+	...
\$2.40	1	+	0	+	...

Panel B: Total Market Demand

Step 2	Step 3	Projection
Total demand across 400 potential customers	Scale up to represent 40 million people	Total market demand
1,200 litres	× 100,000	= 120 million litres
1,100 litres	× 100,000	= 110 million litres
1,000 litres	× 100,000	= 100 million litres
900 litres	× 100,000	= 90 million litres
800 litres	× 100,000	= 80 million litres

Market demand curve

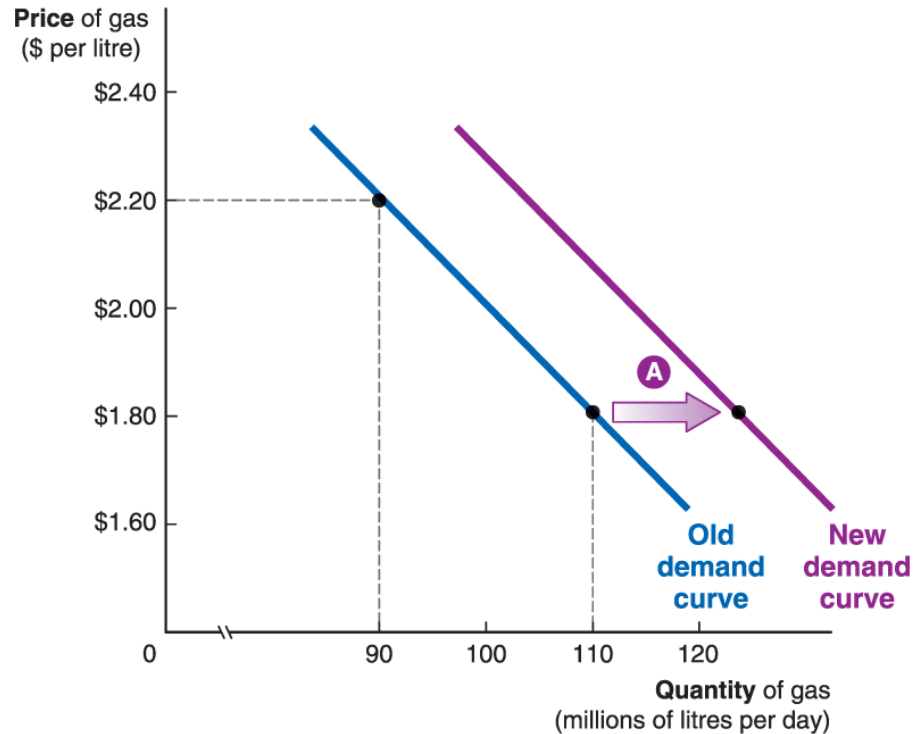


- The **market demand curve** is downward sloping.
- Movements along the market demand curve reflect changes in the quantity demanded of existing customers and the entry/exit of new/old customers from the market.

Shifts in the market demand curve

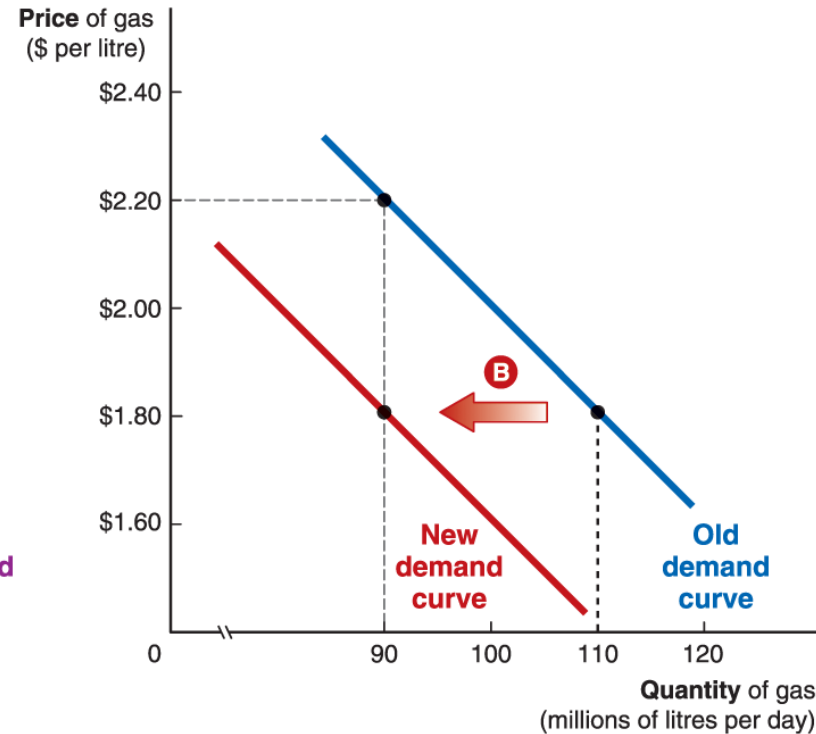
Panel A: An Increase in Demand

- A** An **increase in demand shifts the demand curve to the right**, leading to a higher quantity demanded at each and every price.



Panel B: A Decrease in Demand

- B** A **decrease in demand shifts the demand curve to the left**, leading to a lower quantity demanded at each and every price.



What happens to the **market demand curve** if income tax rates were reduced leaving consumers with more income to spend?

Six factors that shift the demand curve

1. Income

- A higher income allows individuals to buy more of everything.
- A **normal good** is one that we demand more of when our income increases. An **inferior good** is one that we demand less of when our income increases.

2. Preferences

- A marketing campaign that convinces us that we need to wear a certain brand of clothing to be “cool.”

3. Prices of related goods

- A change in the price of a related good.
- When a decrease in the price of a related good increases our demand for a good, the related good is a **complementary good**. When the price decrease of a related good decreases our demand for a good, the related good is a **substitute good**.

Six factors that shift the demand curve

4. Expectations

- Beliefs about whether future prices will be higher or lower affect our demand today.

5. Congestion and network effects

- Our value for a good depends on other people's demand for the same good.
- A **network effect** occurs when a good becomes more valuable to us when more other people are using it. A **congestion effect** occurs when a good becomes less valuable when more other people are using it.

6. The type and number of buyers in the market

- Market demand is the sum of individual demand, an increase in the number of consumers in the market will increase market demand.
- Unlike the first five factors, this factor shifts the market demand curve, but not individual demand curves.

Movements along vs shifts of the demand curve

